Homework Exercise

The objective is to make you practise STICS between the intra-session break, either using JavaStics interface or SticsRPacks, interacting within small groups of 4-5 persons. At the end of session 4, you will present the results of this homework, with 1-2 slides per question (1 to 4) to show the results of your simulations.
To discuss altogether, do not forget to describe accurately your simulations (crop, year, etc.), the calculations you made and the sources you used (if any). Tables and, especially, Figures are encouraged rather than text explanations. You will have 10-15 min per group to present your results on the last afternoon. For those who will choose to do the homework using R, we provided an example of script (called homework_example.R) that may help you (look at it before starting the homework 😉).

<u>Exercise Introduction:</u>
In the semi-arid and arid Mediterranean areas, water scarcity is the main crop-limiting factor. Therefore, the range of feasible crops and their yield is highly limited and variable over years. In some of these rainfed areas, only a few adapted spring crops such as spring barley, spring durum wheat and spring field pea are grown. All these crops are cultivated for grain production. In the current context of relatively high prices for commodities such as these, the crop's choice is essential to maximize the economic income in a farm. In this regard, the STICS model can be of use to simulate the potential and actual yields of the three possible crops for the described area.

1- Using STICS, simulate the potential yield of the three crops (spring barley, spring durum wheat and spring field pea**) for the semi-arid Mediterranean conditions of Raïmat (STICSland).**

You have a folder (STICS_workspace) containing the climate station, climate, soil and initialization files to run the simulations.

A basic USM for each crop is available (spring barley, spring wheat and spring field pea).

Use the standard cropping practices you are provided with (Excel file) to create the .tec file of each crop by modifying the base technical file (ORIGINAL_tec). Months are given for dates of practices, choose the days as you wish.

Choose a year among 2011 to 2020 as you wish and define the climate files for the USM. Since all crops are spring crops requiring a winter soil preparation the preceding year, you must choose two successive years for your simulations (e.g. for a crop harvested in 2011, choose clima_RAIMAT2010 for the climate file of the first year, and clima_RAIMAT2011 for the climate file of the second year). Remember to report the years you have used when presenting your results.

If information needed, see file:///D:/Documents/JavaSTICS-1.5.1-STICS-10.0.0/doc/JavaStics_documentation.html

2- Simulate the actual yields obtained over 10 years, analyse the potential causes of the variability, in particular looking at yield components and environmental stresses; you can focus on one or two species out of the three.

3- Given the low and unstable yields in this area, the STICSland Government has decided to implement an irrigation system to overcome this situation. Irrigation can dramatically improve the yield and its stability. However, both the rate and the distribution are essential factors contributing to the crop response to irrigation. Imagine you have to provide the Government an optimum irrigation rate for one or two of the most important crops in this area. Fixed or automatic irrigation options are also available.

- **Use the STICS model to calculate the yield response to irrigation rate for two of the previous crops and report it as a figure.**
- **Which is the optimum irrigation rate to reach the highest yield on each crop?**

Nb: Account for the fact that irrigation should be allocated during the crop's highest water need, which is typically around the flowering stage for the studied species.

4- Once irrigation is provided to reach the optimum yield, the yield is expected to increase dramatically in these areas. Therefore, crop management should be modified accordingly. Among others, N fertilization is an important factor to take into consideration to satisfy higher-yielding crops.

- **Use the STICS model to calculate the yield response to N fertiliser rate for two of the previous irrigated crops. Which is the optimum nitrogen rate?**

- **In that case, what are the environmental outcomes, as nitrate leaching?** Propose a simulation protocol to find a compromise between production and environment